



OCR SOFTWARE

Optical Character Recognition (OCR) is the process of translation of optically scanned bitmaps of written or printed text into character codes such as ASCII—meaning you scan a newspaper clip (or anything else) and extract the text from it. OCR software is used to convert a scanned document into a document understandable by a word processor.

Such software is not limited to offices; it can even be used at home. As an example, you can scan a form, use OCR to convert it to a document, and fill it in electronically, thus eliminating the need to fill in a new form in case you make a mistake. Or, say you want to send someone a newspaper clip. Instead of cutting up the newspaper and then relying on snail-mail, you could scan the paper and send off the content as an e-mail attachment!

OCR software really does seem amazing the first time you use it—good OCR software will recognise fonts, pictures, formatting, and much more.

Questions to ask the dealer

Does it save in a variety of formats?

Most OCR software save documents in Word format. Apart from this, look for support for saving in other formats such as PDF, HTML and Excel.

Does it support batch processing?

The batch processing feature is very

useful when there are a large number of documents to be scanned. Using this feature, you can just stack up the documents in the ADF (a scanner with an ADF is necessary in this case) instead of sitting in front of the scanner and feeding in one paper at a time.

Interesting Facts

David Shepard, a cryptanalyst at AFSA, the forerunner of the National Security Agency of the USA, along with Dr. Louis Tordella and Harvey Cook, built 'Gismo'—a machine that could convert printed messages to machine language for computer processing, in 1951. Shepard then founded the Intelligent Machines Research Corporation, or IMR, and went on to deliver the world's first OCR systems used in commercial operation. The first IMR scanner was installed at the Readers Digest in 1955. Postal departments around the world have been using such equipment for a very long time now.

Usage tips

❑ The document to be scanned must be free of wrinkles, ears, smudges or creases in order to increase accuracy.

❑ Keep a white paper sheet behind the document to be OCR'd, especially when you are scanning a newspaper or thin paper.

❑ Keep the edges of the document aligned with that of the scanner in

order to increase the OCR's accuracy.

❑ While scanning text documents, use 300 dpi, and scan in line art mode. This will greatly reduce the size of the scanned file without any loss in accuracy. The scanning speed will also be faster.

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❑ If the OCR software allows you to control the threshold (brightness), try using different threshold levels to scan a document that's giving you a high number of errors. You could be able to improve the accuracy.

❑ Keep the glass pane of the scanner clean—this, too, improves the OCR accuracy.

The Future

Advances are being made in the field of OCR, such as the Predictive Optical Word Recognition algorithm from Scansoft Inc. Research is being done on document recognition in which the software will use knowledge of parts of speech and grammar to recognise individual characters. OCR software today offers an accuracy of 90 per cent, but with the introduction of software being able to train, this is expected to improve. Similarly, efforts are underway to improve script and handwritten recognition. Improvements are not only being done on the software algorithms, but also in the hardware aspect, using better quality CCDs in the scanner elements. ■



Points To Ponder

Is it accurate enough for me?

This will depend on the importance of the documents you are OCR-ing: if you will not be OCR-ing critical documents, go for a cheaper OCR software.

Does it have a learning mode?

Many OCR software come with a learning mode. In this, the software learns from its

mistakes during previous OCR sessions, and adapts so as not to repeat the same mistakes.

Is the price justified?

In OCR software, very often, the price doesn't correlate with how good the software is. You'll need to get an informed opinion on this. Try and ask those in the know, and don't rely entirely on what the vendor says.